

PRD & TRD: Satellite-Derived Surface AQI & HCHO Hotspot Tracking

PART 1: PRODUCT REQUIREMENT DOCUMENT (PRD)

1. Executive Summary & Problem Context

Air pollution is a major environmental health hazard responsible for significant excess mortality worldwide. While the Air Quality Index (AQI) simplifies complex multi-pollutant measurements into an actionable metric for public communication and regulatory deployment, physical ambient monitoring stations remain geographically limited. Over 90% of the population across the Indian subcontinent resides further than 100 km from an active physical tracking site. Generating a high-resolution surface AQI derived continuously from satellite observations is vital to achieving complete spatial coverage on a daily scale over India.

Concurrently, Formaldehyde (HCHO) serves as a critical tracer for Volatile Organic Compound (VOC) emissions, playing a dominant role in atmospheric chemistry and tropospheric ozone synthesis. Massive seasonal biomass clearance (crop residue burning and forest fires) releases extensive VOC volumes, drastically worsening downwind pollution exposure. This product defines a precise mechanism to track, map, and analyze the spatio-temporal dynamics of HCHO hotspots over India using satellite observations, prioritizing intensive crop burning intervals.

STRATEGIC GOAL ALLIANCES

- **National Clean Air Programme (NCAP):** Targets a 20–40% drop in particulate matter concentrations across non-attainment regional clusters.
- **UN SDG Target 11.6:** Focuses on reducing urban environmental impacts per capita.
- **Chintan Shivir 2.0:** Focuses on advancing real-time spatial analytics for urban air quality protection, mitigation, and sustainable deployment.

2. Product Vision & Value Proposition

- **Vision:** Build an integrated nationwide geospatial analysis platform that maps near-surface AQI variations daily and exposes statistical VOC emission profiles across India.
- **Value Proposition:** Mitigates physical sensor deficits by delivering a grid-based spatial intelligence tool for regulatory planning, research tracking, and environmental monitoring.

3. Functional Requirements & Core Scope

Module 1: Surface AQI Extraction Engine (Objective-1)

- **FR-1.1:** Continuous automated ingestion pipelines for multi-satellite columns, surface station inputs, and gridded reanalysis models.

- **FR-1.2:** Mathematical downscaling component to translate total columnar metrics into estimated near-surface pollutant concentrations (PM2.5, NO2, SO2, CO, O3).
- **FR-1.3:** Interpolation logic to render daily spatial maps based on the standard CPCB Indian AQI scaling system.

Module 2: HCHO Hotspot & Biomass Tracking (Objective-2)

- **FR-2.1:** Stream real-time thermal anomaly indicators from MODIS and VIIRS satellite arrays.
- **FR-2.2:** Spatial segmentation algorithms to delineate statistically meaningful boundaries of elevated HCHO concentrations.
- **FR-2.3:** Wind trajectory calculations to link active thermal burning nodes directly with downwind plume enhancements.

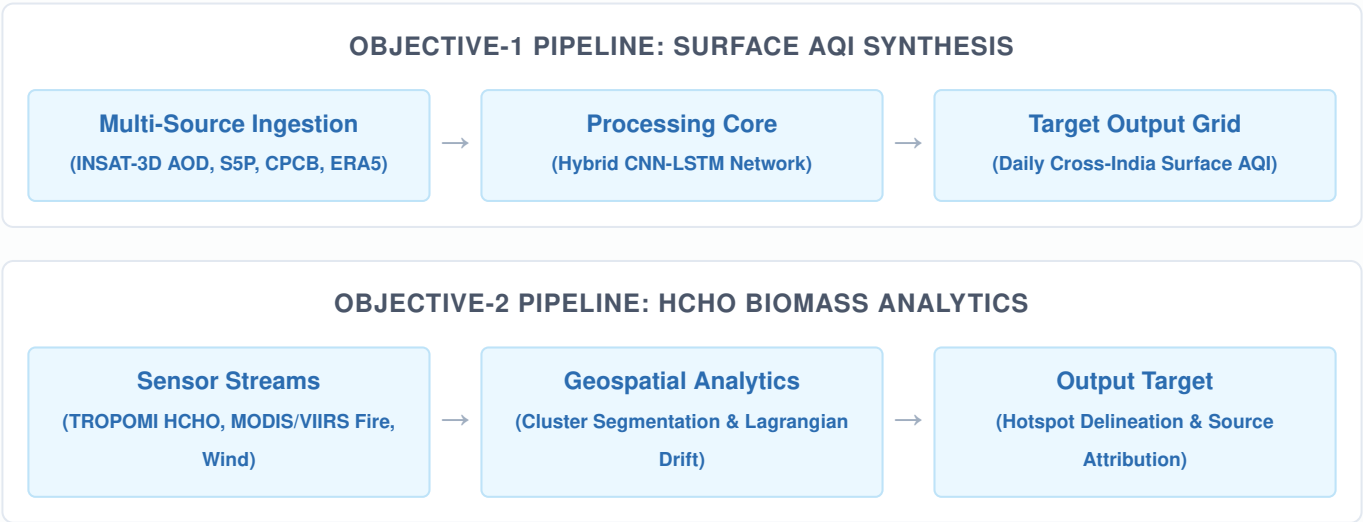
4. Non-Functional Requirements (NFRs)

- **Spatial Resolution:** Outputs must scale to maximize native TROPOMI grids (5.5 km x 3.5 km) and downscaled INSAT-3D profiles.
- **Boundary Masking:** Geoprocessing boundaries must precisely conform to official Indian territorial layouts.
- **System Transparency:** Maintain continuous verification tracking (R^2 , RMSE, MAE) against active physical stations to ensure scientific validation.

PART 2: TECHNICAL REQUIREMENT DOCUMENT (TRD)

1. System Architecture Pipelines

The core computational logic processes data streams via two targeted analytical tracks modeled after the system design layout:

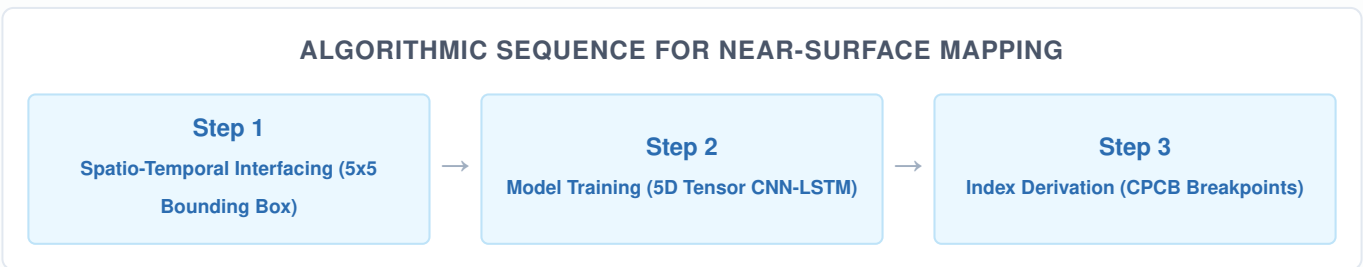


2. Data Ingestion Framework

- **Aerosol Metrics:** Ingest INSAT-3D Aerosol Optical Depth (AOD) products from the MOSDAC portal at a native resolution of 4 km x 4 km (downscaled to 1 km).
- **Trace Gas Columns:** Extract Sentinel-5P TROPOMI vertical columns via Google Earth Engine for HCHO, NO₂, SO₂, CO, and O₃ at 5.5 km x 3.5 km spatial bounds.
- **In-Situ Ground Truth:** Interface directly with CPCB CAAQM network station locations to extract point surface measurements of PM_{2.5}, NO₂, SO₂, CO, and O₃.
- **Thermal Anomaly Data:** Capture active fire pixel counts, timestamps, and Fire Radiative Power (FRP) from MODIS / VIIRS via NASA FIRMS.
- **Meteorological Data:** Load gridded wind vectors, temperature, relative humidity, and planetary boundary layer data at 0.25° grids via ERA5 or IMDAA reanalysis models.

3. Algorithmic Steps & Model Training Workflow

Objective-1 Formulation: Near-Surface Mapping Pipeline



Step 1 (Spatio-Temporal Interfacing): Synchronize multi-sensor satellite grids with meteorological layers. Collocate spatial attributes into an aligned multi-dimensional vector matrix centered directly over real-world CPCB surface stations. Extract a local spatial bounding box patch matrix (e.g., 5 x 5 pixels) around the coordinates to reduce displacement errors caused by wind currents.

Step 2 (Model Training & Tensor Shapes): Structure the input data arrays as 5-dimensional tensors with the shape: *[Batch, Time_Steps, Channels, Height, Width]*. Apply TimeDistributed(Conv2D) filters to extract local spatial features, and feed the output maps into sequential Bidirectional LSTM blocks to capture temporal pollution accumulation trends. Train using a compound objective loss function:

$$Loss = (\alpha \times MSE) + ((1 - \alpha) \times MAE)$$

to avoid overfitting. Implement GroupKFold cross-validation grouped by CPCB Station IDs to ensure geographic generalization.

Step 3 (Index Derivation): Calculate sub-index values across the spatial grid using standard CPCB breakpoints. The absolute grid cell AQI value defaults to the highest sub-index evaluated across the criteria:

$$AQI = \text{Max}(\text{Sub-Index}_{\{PM2.5\}}, \text{Sub-Index}_{\{NO2\}}, \text{Sub-Index}_{\{SO2\}}, \text{Sub-Index}_{\{CO\}}, \text{Sub-Index}_{\{O3\}})$$

Objective-2 Formulation: HCHO Plume & Source Extraction

Step 1 (Baseline Partitioning): Separate normal seasonal background profiles from true fire-driven variations across matching time frames.

Step 2 (Spatial Clustering): Isolate localized regional anomalies by executing Getis-Ord Gi* spatial cluster definitions to capture statistically significant areas of concern with a confidence threshold of Z-score > 2.58 and p-value < 0.01.

Step 3 (Trajectory Evaluation): Map U and V wind vectors at the 850 hPa pressure altitude to model the transport paths of emissions moving downwind from core agricultural burning zones like the Indo-Gangetic Plain.

4. Software Stack & Ecosystem

- **Geospatial Processing:** Google Earth Engine (GEE API), Xarray, Rasterio, and Geopandas.
- **Deep Learning Architecture:** PyTorch or PyTorch Lightning to build and optimize the hybrid CNN-LSTM network layers.
- **Statistical Automation:** Scikit-learn, PySal, and MATLAB for spatial clustering and validation verification.

5. Technical Strategies to Stand Out in the Hackathon

1. **Incorporate Planetary Boundary Layer Height (PBLH):** Surface pollution spikes are heavily influenced by boundary layer shifts. Accounting for dynamic PBLH variations from ERA5 profiles demonstrates a strong understanding of atmospheric physics to the judging panel.

2. **Implement Model Explainability (Spatial SHAP):** Integrate SHAP frameworks into your inference pipelines to explain predictions. Generating feature importance maps helps demonstrate exactly how much specific satellite tracks or meteorological conditions contributed to a predicted spike.
3. **Build a Live Interactive Geospatial Dashboard:** Present findings using a functional Google Earth Engine App or Streamlit dashboard. This lets the jury select a date during the crop residue burning season and instantly visualize downscaled daily surface AQI maps overlaid with active fire locations and wind trajectory lines.